

CE 507: Advanced Concrete Science

Course Code	CE 507
Course Name	Advanced Concrete Science
L-T-P-C	3-0-0-3
Prerequisites	CE 552 - Concrete Technology
Intended for	UG/PG
Distribution	Elective
Approval	9th Senate (May 2016)

Course Contents

- **Module I**
Mix design methods for ordinary and high performance concretes [6 Lectures]
- **Module II**
Performance in fresh and hardened states – Rheology, Stress-Strain characteristics, Creep and Shrinkage. [6 Lectures]
- **Pore structure of cement based materials**
Types and genesis of pores, Methods of measurement. [4 Lectures]
- **Transport processes in concrete**
Mechanisms of Water, Gas and Salt transport; Transport models; Tests for the assessment of transport characteristics. [6 Lectures]
- **Degradation mechanisms**
Carbonation, Sulphate attack, Freeze-Thaw, Alkali-Silica reaction. [6 Lectures]
- **Reinforcement corrosion**
Mechanisms of initiation and propagation, Electrochemical aspects, Monitoring, controlling and prevention, service-life prediction. [6 Lectures]
- **Design for Durability**
Service-life, Prescriptive, Performance and Model-based approaches. [6 Lectures]

Text Books

1. Aligizaki, K. K., **Pore structure of cement-based materials: testing, interpretation and requirements**, Taylor & Francis, New York, 2005.
2. Richardson, M., **Fundamentals of durable reinforced concrete**, Spon Press, London and New York, 2003.
3. Hall, C. and Hoff, W.D., **Water transport in brick, stone and concrete**, CRC Press, Florida, 2011.

References

1. Sarja, A. and Vesikari, E., **Durability design of concrete structures-Report of RILEM technical committee 130-CSL**, E&FN Spon, London, 2004.
2. Cerny, R. and Rovnanikova, P., **Transport processes in concrete**, Spon Press, USA and Canada, 2002.
3. Claisse, P.A., **Transport properties of concrete: measurement and applications**, Woodhead Publishing, UK, 2014.
4. Bertolini, L., Elsener, B., Pedersen, P., Redaelli, E. and Polder, R.B., **Corrosion of steel in concrete: prevention, diagnosis, repair**, Wiley-VCH Verlag GmbH & Co. KGaA, Germany, 2013.
5. J. D. West and F. K. Levy, **A management guide to PERT/CPM with GERT/PDM/DCPM**, 1979.